

UChicago REU Notes Week 2

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1 Week 2

1.1 Intro

For this portion, I want to start with some basic definitions covered by professor May in his office, about the definition of spectra / prespectra / generalized cohomology theory. I'll probably skip some of the details in the intro to make sure I can keep on track. (Also, I probably involve quite a lot of errors in between...I'll try to fix them)

1.1.1 Brown Representability Theorem, and Generalized Cohomology

Let $h\mathcal{C}$ denotes the homotopy category of based spaces (that admits a CW structure, as on them weak equivalences and homotopy equivalence are the same), then a functor $F : h\mathcal{C}^{\text{op}} \rightarrow \mathbf{Ab}$ is *representable*, if there exists a based space $Y \in h\mathcal{C}$, such that $F \cong [-, Y]$.

Example 1.1. The *reduced cohomology functor* $\tilde{H}^n(-; A)$ is one, for all $n \geq 0$ and a fixed abelian group A (as coefficient). Let $K(A, n)$ be the *Eilenberg-MacLane Space* of abelian group A and index $n \geq 0$ (i.e. the n th homotopy group $\pi_n(K(A, n)) = A$), then the following group isomorphism holds:

$$\tilde{H}^n(X; A) \cong [X, K(A, n)]$$

Using more adjunction property, since $\pi_n(A, n) = A = \pi_{n+1}(K(A, n+1)) = [S^{n+1}, K(A, n+1)] = [S^n \wedge S^1, K(A, n+1)] \cong [S^n, \Omega K(A, n+1)] = \pi_n(\Omega K(A, n+1))$, there is a homotopy equivalence $K(A, n) \xrightarrow{\cong} \Omega K(A, n+1)$.

(Yeah, need to understand more about the details)

So, the regular cohomology we see on spaces, are one type of representable functors.

The classification of all such functors is given by *Brown Representability Theorem*:

Theorem 1.2. Suppose the functor $F : h\mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ satisfies:

- $F(\bigvee_{\alpha} X_{\alpha}) \cong \prod_{\alpha} F(X_{\alpha})$ (i.e. maps coproduct to product).
- F maps homotopy pushouts (I believe this is the reduced mapping cylinder?) to weak pullbacks (omit the uniqueness of maps). This condition is equivalent to Mayer-Vietoris axiom (which seems like a weaker form of the gluing axiom in sheaf theory, and restrict to CW subcomplexes instead of open sets).

Then, F is representable by some CW complex C , or $F \cong [-, C]$.

The converse for the theorem also holds, as $[-, C]$ also satisfies the above axioms (need more details here).

Here is a definition of generalized cohomology:

Definition 1.3. A *generalized cohomology*, is a sequence of spaces $\{E_n\}_{n \geq 0}$, with homotopy equivalences $\{\epsilon_n : E_n \xrightarrow{\sim} \Omega E_{n+1}\}_{n \geq 0}$.

In a more explicit form, $E^n(X) := [X, E_n]$ has all $n \geq 0$ satisfies $E^n(X) = [X, E_n] \cong [X, \Omega E_{n+1}]$, which is naturally a group (and an abelian group for $n \geq 2$, due to the abelian group structure on $[X, \Omega^m Y]$ for $m \geq 2$).

I remembered the motivation is that, by Brown Representability Theorem, if a sequence of representable functors $E^* : h\mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ satisfy similar properties to ordinary cohomology (for instance, the connecting morphism $E^n \rightarrow E^{n+1}$, and excision axiom?), then Brown Representability Theorem guarantees it must be the above form. So, they're precisely all the generalized cohomology.

1.1.2 Ω -Prespectra, Prespectra, and Spectra

When talking about generalized cohomology, we have a sequence of based spaces $\{E_n\}_{n \geq 0}$, together with homotopy equivalences $\epsilon_n : E_n \xrightarrow{\sim} \Omega E_{n+1}$, which is a prototype of what's called *Ω -Prespectra*.

In general, fixing a based space $X \in h\mathcal{C}$, the sequence $\{\Sigma^n X\}_{n \geq 0}$ has a collection of natural maps $\{\text{id}_n : \Sigma(\Sigma^n X) \rightarrow \Sigma^{n+1} X\}_{n \geq 0}$, which the adjunction (Σ, Ω) guarantees a collection of maps $\{\eta_n : \Sigma^n X \rightarrow \Omega(\Sigma^{n+1} X)\}_{n \geq 0}$. In general these maps are not homotopy equivalence, so these type of sequences are a type of *Prespectra*, called *suspension prespectra*. More generally, *Prespectra* is a sequence of spaces $\{X_n\}_{n \geq 0}$, with morphisms $\{\mu_n : X_n \rightarrow \Omega X_{n+1}\}_{n \geq 0}$.

For the strongest form possible, if such map $\epsilon_n : E_n \rightarrow \Omega E_{n+1}$ is a homeomorphism, then these are called *Spectra*.

Professor May restricts to three specific categories:

- Let $\mathcal{P}$ denotes the *category of prespectra*.
- Let $\mathcal{Q}$ denotes the *category of inclusion prespectra*, meaning each defining morphism $\mu_n : X_n \rightarrow \Omega X_{n+1}$ is an inclusion.
- Let $\mathcal{S}$ denotes the *category of spectra*.

For each category, we require the morphism $\phi : X. \rightarrow T.$ to be a sequence of based maps $\phi_n : X_n \rightarrow T_n$, such that the following diagram commutes:

$$\begin{array}{ccc} \Sigma X_n & \xrightarrow{\Sigma \phi_n} & \Sigma T_n \\ \mu'_{X_n} \downarrow & & \downarrow \mu'_{T_n} \\ X_{n+1} & \xrightarrow[\phi_{n+1}]{} & T_{n+1} \end{array}$$

where the map $\Sigma X_n \rightarrow X_{n+1}$ is based on the adjoint property $[\Sigma X_n, X_{n+1}] \cong [X_n, \Omega X_{n+1}]$. Here professor May had talked about how the prototype of the morphism only requires it to commute up to homotopy, but turns out it's too weak as a condition.

With these morphisms, there is an obvious forgetful functor $\mathcal{S} \rightarrow \mathcal{Q} \rightarrow \mathcal{P}$. It seems to satisfy all the conditions for Freyd's Adjoint Functor Theorem to hold, both of the forgetful functors have a left adjoint.

Here, the left adjoint of $\mathcal{Q} \rightarrow \mathcal{P}$ is hard to construct, yet there is a clear construction for the left adjoint of $\mathcal{S} \rightarrow \mathcal{Q}$, denoted as $L : \mathcal{Q} \rightarrow \mathcal{S}$, by $T. \mapsto (LT).$, where $(LT)_n := \text{colim}_q \Omega^{n+q} T_q$ (due to the inclusion $T_q \rightarrow \Omega T_{q+1}$, it induces $\Omega^{n+q} T_q \rightarrow \Omega^{n+(q+1)} T_{q+1}$, hence having a sequence of direct system).

Professor May had claimed the homeomorphism $(LT)_n \cong \Omega(LT)_{n+1}$ due to the fact that Ω commutes with colimit (I do need to work out the detail, as normally Ω as a right adjoint of Σ would be thought of preserving limits instead of colimits).

1.2 Review: Adjoint Functors

This part is converging to the preparation of the theory of monads :) (as professor May had mentioned it quite often as an essential tool to understand operad, so I decided to do related review).

Definition 1.4. Let $\mathcal{C}, \mathcal{D}$ be two categories. Two opposite functors $F : \mathcal{C} \rightarrow \mathcal{D}, \mathcal{C} \leftarrow \mathcal{D} : G$ are *adjoint pairs* - in particular, F is a *left adjoint* of G , and G is a *right adjoint* of F - if there is a natural isomorphism

$$\text{Hom}_{\mathcal{D}}(F(X), Y) \cong \text{Hom}_{\mathcal{C}}(X, G(Y))$$

For any $X \in \mathcal{C}$ and $Y \in \mathcal{D}$.

Notation wise, it's denoted as (F, G) or $F \dashv G$.

More formally, it's a natural isomorphism $(-)^\sharp : \text{Hom}_{\mathcal{D}}(F(-), -) \xrightarrow{\sim} \text{Hom}_{\mathcal{C}}(-, G(-))$ of bifunctors $\mathcal{C}^{\text{op}} \times \mathcal{D} \rightarrow \text{Set}$ (here, we also denote $(-)^\flat$ as the inverse transformation).

This means, given morphism $f : X \rightarrow X'$ in $\mathcal{C}$, and morphism $g : Y \rightarrow Y'$ in $\mathcal{D}$, we have diagrams as follow:

$$\begin{array}{ccc} \text{Hom}_{\mathcal{D}}(F(X'), Y) & \xrightarrow{\sim} & \text{Hom}_{\mathcal{C}}(X', G(Y)) , \\ \downarrow \circ Ff & & \downarrow \circ f \\ \text{Hom}_{\mathcal{D}}(F(X), Y) & \xrightarrow{\sim} & \text{Hom}_{\mathcal{C}}(X, G(Y)) \end{array} \quad \begin{array}{ccc} \text{Hom}_{\mathcal{D}}(F(X), Y) & \xrightarrow{\sim} & \text{Hom}_{\mathcal{C}}(X, G(Y)) \\ \downarrow g \circ - & & \downarrow Gg \circ - \\ \text{Hom}_{\mathcal{D}}(F(X), Y') & \xrightarrow{\sim} & \text{Hom}_{\mathcal{C}}(F(X), Y') \end{array}$$

The first diagram insists that any morphism $h : F(X') \rightarrow Y$ in $\mathcal{D}$ satisfies $(h \circ Ff)^\sharp = h^\sharp \circ f$ in $\mathcal{C}$. The second diagram insists that any morphism $\ell : F(X) \rightarrow Y$ satisfies $(g \circ \ell)^\sharp = Gg \circ \ell^\sharp$. Or, the following diagram holds in $\mathcal{C}$:

$$\begin{array}{ccc} X' & & \\ f \downarrow & \searrow (h \circ f)^\sharp & \\ X & \xrightarrow{h^\sharp} & G(Y) \end{array} \quad , \quad \begin{array}{ccc} X & \xrightarrow{\ell^\sharp} & G(Y) \\ & \searrow (g \circ \ell)^\sharp & \downarrow Gg \\ & & G(Y') \end{array}$$

If use the inverse transformation $(-)^\flat$, the above can also be phrased as: Any morphism $\tilde{h} : X' \rightarrow G(Y)$ satisfies $(\tilde{h} \circ f)^\flat = (\tilde{h})^\flat \circ Ff$, and any morphism $\tilde{\ell} : X \rightarrow G(Y)$ satisfies $(Gg \circ \tilde{\ell})^\flat = g \circ (\tilde{\ell})^\flat$ in $\mathcal{D}$.

A property of adjoint functor is it connects the commutativity of certain diagrams between the two categories:

Lemma 1.5. Fix arbitrary $f : X \rightarrow X'$ in $\mathcal{C}$, $g : Y \rightarrow Y'$ in $\mathcal{D}$, and two morphisms $h : F(X) \rightarrow Y$, $h' : F(X') \rightarrow Y'$ in $\mathcal{D}$. The commutativity of following two diagrams are equivalent:

$$\begin{array}{ccc} F(X) & \xrightarrow{h} & Y \\ Ff \downarrow & & \downarrow g \\ F(X') & \xrightarrow{h'} & Y' \end{array} \quad \text{in } \mathcal{D}, \quad \begin{array}{ccc} X & \xrightarrow{h^\sharp} & G(Y) \\ f \downarrow & & \downarrow Gg \\ X' & \xrightarrow{h'^\sharp} & G(Y') \end{array} \quad \text{in } \mathcal{C}$$

Proof. As a remark, $(-)^\sharp$ is a bijection on each homset, so two morphisms $F(X) \rightarrow Y'$ are equal iff after applying $(-)^\sharp$ they're equal as morphisms $X \rightarrow G(Y')$.

Using the notation from above, we see the following equivalence:

$$g \circ h = h' \circ Ff \iff (g \circ h)^\sharp = (h' \circ Ff)^\sharp \iff Gg \circ h^\sharp = h'^\sharp \circ f$$

Where, the second equivalence is the property from above. The left and right most equalities represent the two commutative diagrams respectively, hence showing the equivalence of commutativity. $\square$

Given the adjoint properties, we have the following two natural isomorphisms, given any $X \in \mathcal{C}$ and $Y \in \mathcal{D}$:

$$(_)^\sharp : \text{Hom}_{\mathcal{D}}(F(X), F(X)) \xrightarrow{\sim} \text{Hom}_{\mathcal{C}}(X, GF(X)), \quad \text{Hom}_{\mathcal{D}}(FG(Y), Y) \xleftarrow{\sim} \text{Hom}_{\mathcal{C}}(G(Y), G(Y)) : (_)^\flat$$

Hence, we can define a collection of maps $\eta_X := (\text{id}_{F(X)})^\sharp : X \rightarrow GF(X)$, and $\epsilon_Y := (\text{id}_{G(Y)})^\flat : FG(Y) \rightarrow Y$.

Proposition 1.6. *The two collections form natural transformations $\eta : \text{Id}_{\mathcal{C}} \rightarrow GF$, and $\epsilon : FG \rightarrow \text{Id}_{\mathcal{D}}$.*

Proof. Here, fix arbitrary $f : X \rightarrow X'$ in $\mathcal{C}$, and $g : Y \rightarrow Y'$ in $\mathcal{D}$. Using **Lemma 1.5**, the following two diagrams have equivalent commutativity:

$$\begin{array}{ccc} F(X) & \xrightarrow{\text{id}_{F(X)}} & F(X) \text{ in } \mathcal{D}, \\ Ff \downarrow & & \downarrow Ff \\ F(X) & \xrightarrow{\text{id}_{F(X')}} & F(X') \end{array} \quad \begin{array}{ccc} X & \xrightarrow{(\text{id}_{F(X)})^\sharp = \eta_X} & GF(X) \text{ in } \mathcal{C} \\ f \downarrow & & \downarrow GFf \\ X' & \xrightarrow{(\text{id}_{F(X')})^\sharp = \eta_{X'}} & GF(X') \end{array}$$

Where, the left side obviously commutes, so the right side commutes, showing naturality of η .

Dually, since $(\epsilon_Y)^\sharp = \text{id}_{G(Y)}$ by the inverse property, we also have the equivalence of the following two diagrams:

$$\begin{array}{ccc} FG(Y) & \xrightarrow{\epsilon_Y} & Y \text{ in } \mathcal{D}, \\ FGg \downarrow & & \downarrow g \\ FG(Y) & \xrightarrow{\epsilon_{Y'}} & Y' \end{array} \quad \begin{array}{ccc} G(Y) & \xrightarrow{(\epsilon_Y)^\sharp = \text{id}_{G(Y)}} & G(Y) \text{ in } \mathcal{C} \\ f \downarrow & & \downarrow Gg \\ G(Y') & \xrightarrow{(\epsilon_{Y'})^\sharp = \text{id}_{G(Y')}} & G(Y') \end{array}$$

Where, the right side obviously commutes, so the left side commutes, showing the naturality of ϵ . □

These natural transformations are named as follow:

Definition 1.7. Given the adjoint pairs $F \dashv G$, the natural transformation $\eta : \text{Id}_{\mathcal{C}} \rightarrow GF$ is called the *unit transformation*, while $\epsilon : FG \rightarrow \text{Id}_{\mathcal{D}}$ is called the *counit transformation*.

They satisfy what's called the *Triangle Identity* as follow:

Proposition 1.8. *The following two diagrams of natural transformation holds:*

$$\begin{array}{ccc} F & \xrightarrow{F\eta} & FGF \text{ in } \mathcal{D}^{\mathcal{C}}, \\ \text{id}_F \searrow & & \downarrow \epsilon_F \\ & & F \end{array} \quad \begin{array}{ccc} G & \xrightarrow{\eta_G} & GFG \text{ in } \mathcal{C}^{\mathcal{D}} \\ \text{id}_G \searrow & & \downarrow G\epsilon \\ & & G \end{array}$$

Here, $\mathcal{D}^{\mathcal{C}}$ denotes the *functor categories* of all functors $\mathcal{C} \rightarrow \mathcal{D}$, with morphisms being natural transformations between functors.

Proof. Given arbitrary $X \in \mathcal{C}$, the goal is to show the horizontal morphisms of the following diagram are identities:

$$F(X) \xrightarrow{F\eta_X} FGF(X) \xrightarrow{\epsilon_{F(X)}} F(X)$$

Using out notation and the naturality, we see the following equality of morphisms after applying $(_)^\sharp$:

$$(\epsilon_{F(X)} \circ F\eta_X)^\sharp = (\epsilon_{F(X)})^\sharp \circ \eta_X = \text{id}_{GF(X)} \circ \eta_X = \eta_X : GF(X) \rightarrow X$$

But, recall that $\eta_X := (\text{id}_{F(X)})^\sharp$, using the bijectivity of $(-)^\sharp$, this enforces $\epsilon_{F(X)} \circ F\eta_X = \text{id}_{F(X)}$, showing $\epsilon_F \circ F\eta = \text{id}_F$ as natural transformations.

Dually, given arbitrary $Y \in \mathcal{D}$, the goal is again to show the following morphism is identity:

$$G(Y) \xrightarrow{\eta_{G(Y)}} GFG(Y) \xrightarrow{G\epsilon_Y} G(Y)$$

Similarly, after applying $(-)^\flat$ we get:

$$(G\epsilon_Y \circ \eta_{G(Y)})^\flat = \epsilon_Y \circ (\eta_{G(Y)})^\flat = \epsilon_Y \circ \text{id}_{FG(Y)} = \epsilon_Y$$

The second equality relies on the fact $(\text{id}_{F(X)})^\sharp = \eta_X$. Which, since $(-)^\flat$ is again a bijection, and $\epsilon_Y := (\text{id}_{G(Y)})^\flat$, we see $G\epsilon_Y \circ \eta_{G(Y)} = \text{id}_{G(Y)}$, showing $G\epsilon \circ \eta_G = \text{id}_G$ as natural transformations. $\square$

This will be an important result when classifying monads. We also want to say the converse of Triangle Identity holds, that it generates an adjoint pair:

Lemma 1.9. *Suppose $F : \mathcal{C} \rightarrow \mathcal{D}$ and $\mathcal{C} \leftarrow \mathcal{D} : G$ are two opposite functors, together with natural transformations $\eta : \text{Id}_{\mathcal{C}} \rightarrow GF$ and $\epsilon : FG \rightarrow \text{Id}_{\mathcal{D}}$ such that Triangle Identity holds:*

$$\begin{array}{ccc} F & \xrightarrow{F\eta} & FGF \text{ in } \mathcal{D}^{\mathcal{C}}, \\ & \searrow \text{id}_F & \downarrow \epsilon_F \\ & & F \end{array} \quad \begin{array}{ccc} G & \xrightarrow{\eta_G} & GFG \text{ in } \mathcal{C}^{\mathcal{D}} \\ & \searrow \text{id}_G & \downarrow G\epsilon \\ & & F \end{array}$$

then $F \dashv G$ is an adjoint pair.

Proof. Given any $X \in \mathcal{C}$ (associated to $\eta_X : X \rightarrow GF(X)$) and $Y \in \mathcal{D}$ (associated to $\epsilon_Y : FG(Y) \rightarrow Y$), for any morphism $h : F(X) \rightarrow Y$, we have a corresponding morphism $Gh : GF(X) \rightarrow G(Y)$. This defines the following map:

$$(-)^\sharp : \text{Hom}_{\mathcal{D}}(F(X), Y) \rightarrow \text{Hom}_{\mathcal{C}}(X, G(Y)), \quad h^\sharp := Gh \circ \eta_X$$

Similarly, for any morphism $\ell : X \rightarrow G(Y)$, we have a corresponding morphism $F\ell : F(X) \rightarrow FG(Y)$. This again defines the following map:

$$\text{Hom}_{\mathcal{D}}(F(X), Y) \leftarrow \text{Hom}_{\mathcal{C}}(X, G(Y)) : (-)^\flat, \quad \ell^\flat := \epsilon_Y \circ F\ell$$

To show they're mutual inverses, we expand the following:

$$(h^\sharp)^\flat = \epsilon_Y \circ Fh^\sharp = \epsilon_Y \circ F(Gh \circ \eta_X) = \epsilon_Y \circ FGh \circ F\eta_X$$

Expanding the right hand side in terms of diagram (includes the triangle identity and naturality of ϵ), we get:

$$\begin{array}{ccccc} & & \text{id}_{F(X)} & & \\ & \swarrow & & \searrow & \\ F(X) & \xrightarrow{F\eta_X} & FGF(X) & \xrightarrow{\epsilon_{F(X)}} & F(X) \\ & & \downarrow FGh & & \downarrow h \\ & & FG(Y) & \xrightarrow{\epsilon_Y} & Y \end{array}$$

So, we see $(h^\sharp)^\flat = \epsilon_Y \circ FGh \circ F\eta_X = h$ by the above commutative diagram.

Expand the dual diagram with the other triangle identity and naturality of η , we'll get that $(\ell^\flat)^\sharp = \ell$ instead, showing the two are mutual inverses. $\square$

1.3 Monads and Monad Algebra